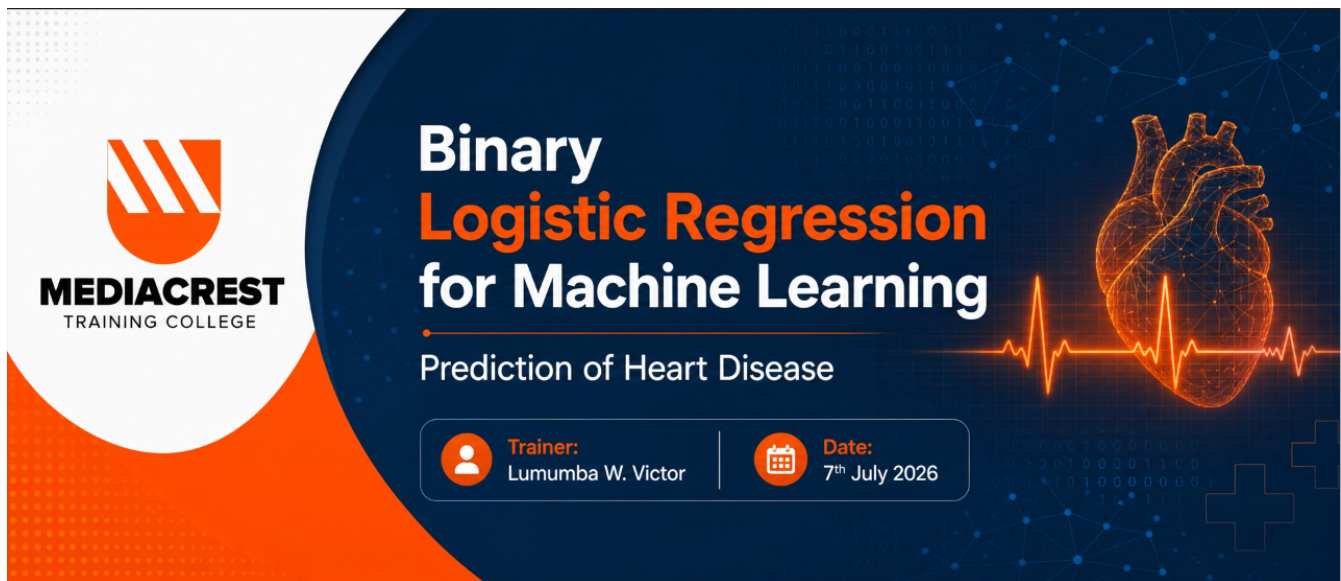




LOAD THE NEEDED LIBRARIES

In [1]:

```
### Load the libraries for data importation, numeric manipulation and visualization
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import statsmodels.api as sm

### Load the libraries for data preprocessing and data partitioning
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report, roc

import warnings
warnings.filterwarnings("ignore")
```

LOAD THE DATASET

In [3]:

```
df = pd.read_csv('Heart.csv', na_values=['NA', '?', ''])
df.head()
```

Out[3]:

	Age	Sex	ChestPain	RestBP	Chol	Fbs	RestECG	MaxHR	ExAng	Oldpeak	Slope	Ca	1
0	63	1	typical	145	233	1	2	150	0	2.3	3	0.0	f
1	67	1	asymptomatic	160	286	0	2	108	1	1.5	2	3.0	noi
2	67	1	asymptomatic	120	229	0	2	129	1	2.6	2	2.0	revers
3	37	1	nonanginal	130	250	0	0	187	0	3.5	3	0.0	noi
4	41	0	nontypical	130	204	0	2	172	0	1.4	1	0.0	noi

Drop AHD Since it is a String Version of the Target Variable HD

In [4]:

```
df = df.drop(columns = ['AHD'])  
df.head()
```

Out[4]:

	Age	Sex	ChestPain	RestBP	Chol	Fbs	RestECG	MaxHR	ExAng	Oldpeak	Slope	Ca	1
0	63	1	typical	145	233	1	2	150	0	2.3	3	0.0	f
1	67	1	asymptomatic	160	286	0	2	108	1	1.5	2	3.0	no
2	67	1	asymptomatic	120	229	0	2	129	1	2.6	2	2.0	revers
3	37	1	nonanginal	130	250	0	0	187	0	3.5	3	0.0	no
4	41	0	nontypical	130	204	0	2	172	0	1.4	1	0.0	no

Handling Missing Values

In [5]:

```
df.info()
```

```
<class 'pandas.DataFrame'>  
RangeIndex: 303 entries, 0 to 302  
Data columns (total 14 columns):  
#   Column      Non-Null Count  Dtype  
---  -  
0   Age         303 non-null    int64  
1   Sex         303 non-null    int64  
2   ChestPain   303 non-null    str  
3   RestBP      303 non-null    int64  
4   Chol        303 non-null    int64  
5   Fbs         303 non-null    int64  
6   RestECG     303 non-null    int64  
7   MaxHR       303 non-null    int64  
8   ExAng       303 non-null    int64  
9   Oldpeak     303 non-null    float64  
10  Slope       303 non-null    int64  
11  Ca          299 non-null    float64  
12  Thal        301 non-null    str  
13  HD          303 non-null    int64  
dtypes: float64(2), int64(10), str(2)  
memory usage: 38.7 KB
```

Check the Sum of Missing Values per Variable

In [6]:

```
df.isna().sum()
```

Out[6]:

```
Age      0  
Sex      0  
ChestPain 0  
RestBP   0  
Chol     0
```

```
Fbs          0
RestECG      0
MaxHR        0
ExAng        0
Oldpeak      0
Slope        0
Ca           4
Thal         2
HD           0
dtype: int64
```

Data Imputation (Median and Mode)

In [7]:

```
df['Ca'] = df['Ca'].fillna(df['Ca'].median())
df['Thal'] = df['Thal'].fillna(df['Thal'].mode()[0])
```

In [8]:

```
df.isna().sum()
```

Out[8]:

```
Age          0
Sex          0
ChestPain    0
RestBP       0
Chol         0
Fbs          0
RestECG      0
MaxHR        0
ExAng        0
Oldpeak      0
Slope        0
Ca           0
Thal         0
HD           0
dtype: int64
```

In [9]:

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 303 entries, 0 to 302
Data columns (total 14 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   Age         303 non-null   int64
 1   Sex         303 non-null   int64
 2   ChestPain   303 non-null   str
 3   RestBP      303 non-null   int64
 4   Chol        303 non-null   int64
 5   Fbs         303 non-null   int64
 6   RestECG     303 non-null   int64
 7   MaxHR       303 non-null   int64
 8   ExAng       303 non-null   int64
 9   Oldpeak     303 non-null   float64
10   Slope       303 non-null   int64
11   Ca          303 non-null   float64
12   Thal        303 non-null   str
13   HD          303 non-null   int64
```

dtypes: float64(2), int64(10), str(2)
memory usage: 38.7 KB

Seperate X-Matrix of Predictor and y-Predictor Variable

In [10]:

```
X = df.drop(columns = ['HD'])
X.head()
```

Out[10]:

	Age	Sex	ChestPain	RestBP	Chol	Fbs	RestECG	MaxHR	ExAng	Oldpeak	Slope	Ca	1
0	63	1	typical	145	233	1	2	150	0	2.3	3	0.0	f
1	67	1	asymptomatic	160	286	0	2	108	1	1.5	2	3.0	noi
2	67	1	asymptomatic	120	229	0	2	129	1	2.6	2	2.0	revers
3	37	1	nonanginal	130	250	0	0	187	0	3.5	3	0.0	noi
4	41	0	nontypical	130	204	0	2	172	0	1.4	1	0.0	noi

In [11]:

```
y = df['HD']
y.head()
```

Out[11]:

```
0    0
1    1
2    1
3    0
4    0
Name: HD, dtype: int64
```

Get Dummies for Character Variables

In [12]:

```
df['ChestPain'].value_counts()
```

Out[12]:

```
ChestPain
asymptomatic    144
nonanginal      86
nontypical      50
typical         23
Name: count, dtype: int64
```

In [13]:

```
df['Thal'].value_counts()
```

Out[13]:

```
Thal
normal    168
reversable 117
fixed     18
Name: count, dtype: int64
```

In [14]:

```
X = pd.get_dummies(X, drop_first = True)
X.head()
```

Out[14]:

	Age	Sex	RestBP	Chol	Fbs	RestECG	MaxHR	ExAng	Oldpeak	Slope	Ca	ChestPain_nonangina
0	63	1	145	233	1	2	150	0	2.3	3	0.0	False
1	67	1	160	286	0	2	108	1	1.5	2	3.0	False
2	67	1	120	229	0	2	129	1	2.6	2	2.0	False
3	37	1	130	250	0	0	187	0	3.5	3	0.0	True
4	41	0	130	204	0	2	172	0	1.4	1	0.0	False

Convert the Boolean Data Types to Binary

```
In [15]:
bool_cols = X.select_dtypes(include = 'bool').columns
X[bool_cols] = X[bool_cols].astype(int)
X.head()
```

Out[15]:

	Age	Sex	RestBP	Chol	Fbs	RestECG	MaxHR	ExAng	Oldpeak	Slope	Ca	ChestPain_nonangina
0	63	1	145	233	1	2	150	0	2.3	3	0.0	
1	67	1	160	286	0	2	108	1	1.5	2	3.0	
2	67	1	120	229	0	2	129	1	2.6	2	2.0	
3	37	1	130	250	0	0	187	0	3.5	3	0.0	
4	41	0	130	204	0	2	172	0	1.4	1	0.0	

EXPLORATORY DATA ANALYSIS (EDA)

```
In [ ]:
In [ ]:
In [ ]:
In [ ]:
In [ ]:
In [ ]:
```

In []:

In []: